Corporate Finance

Lecture 12: Options and Risk Management

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Plan

A. Futures & forwards: simple financial derivatives

B. Option basics (Chap. 20.1)

C. Option payoffs at expiration (Chap. 20.2)

Introduction

- A company is exposed to many different risks:
  - Changes in demand
  - Changes in the cost of commodities
  - Loss of key employees
  - New competitors entering the market
  - Operational risks, etc.

Introduction

- Important question for the company:
  - Should investors support these risks? or
  - Should the company reduce/manage the risks?
- Managers have different tools to reduce risks:
  - Prevention.
  - Hedging:
    - with insurance contracts.
    - in financial markets.

Insurance vs. hedging with derivatives

- Some risks are easy to insure:
  - Accidents, illness, fire.
- For other risks it is hard or impossible to find insurance:
  - Climate risk,
  - Risk of natural disasters,
  - Cost of commodities.
- In particular, it is usually easier to hedge a "non diversifiable" or "hard to diversify" risk with a financial derivative than with an insurance contract.

Financial derivatives

- The value of financial derivatives depends on another value (the underlying).
- The underlying is mostly the price of a financial security.
- Example:
  - Credit default swaps (CDS),
  - Swaps,
  - Futures/forwards,
  - Options.

A. Futures & forwards: simple financial derivatives

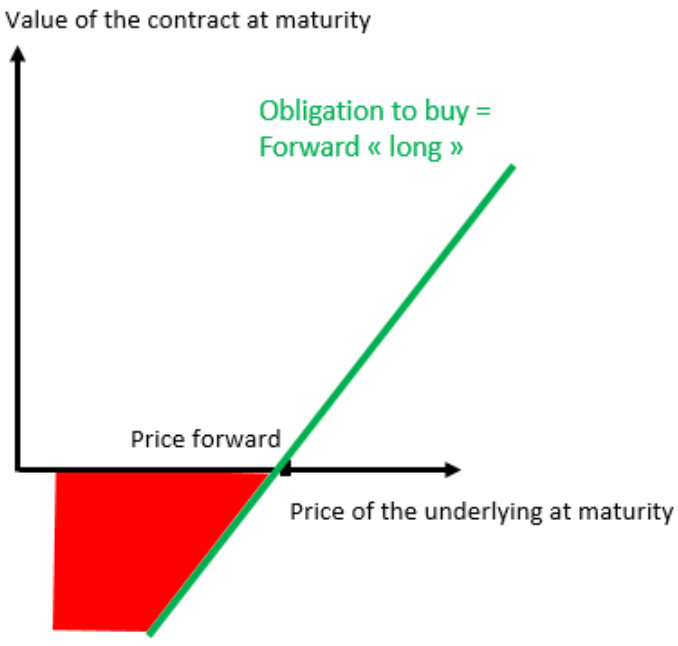
- A forward is a legal agreement between a seller and a buyer:
  - The buyer agrees to buy something at a specified price and a specified date,
  - The seller agrees to sell something at a specified price and a specified date.
- A market for standardised forward contracts equipped with a clearing house is called a futures market.
  - For futures there are margin requirements and the clearing house is the counterparty so that there is no credit risk in the futures market.

Forward/futures: terminology

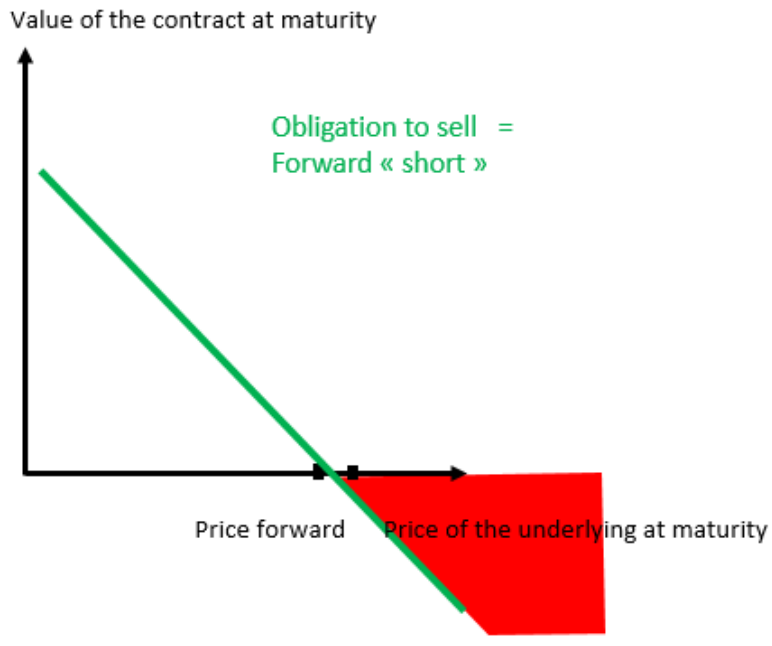
"Spot" market	**"Futures/forward" market**
Price	Futures/forward price
Today	Expiration date or transaction date
Something	Underlying
Buyer	Party which takes the long position in the contract
Seller	Party which takes the short position in the contract

The value of a forward contract

The value of the forward contract at maturity depends on the spot price at maturity:



If at maturity the price is below the forward price, the buyer could have bought cheaper without the forward contract. He thus experiences a loss.



If at maturity the price is larger than the forward price, the seller has to sell for a price below the market price. He thus experiences a loss.

Closing a position

- The buyer makes a profit if the price of the underlying increases.
- The seller makes a profit if the price of the underlying decreases.
- The futures contract can be held until expiry. The expiration dates are usually in March, June, September, and December.
- The position can be closed before the expiration date: an investor with a long position can take a short position on the same contract.

At the maturity of the contract

- The settlement can be made in cash (**cash settlement**).
  - Example: stock index.
- Or by the delivery of the underlying asset, real or financial (**physical delivery**).
  - Example: bonds, commodities.

B. Option basics

- The major drawback of forward and future contracts is the possibility to experience a negative payoff.
- A forward contract that does not force the two parties to execute is called an "option."
- Thus, an option corresponds to a forward contract which only has a non-negative payoff (positive or zero).
- Because an option is a bet in which you cannot lose, you need to pay the price of the option to enter the bet.

Call & put options

- **Call**: gives the owner the right, but not the obligation, to buy a fixed quantity of the underlying asset at a certain price at (or before) a certain date.
  - Exercise price or strike price: The price at which an option holder buys or sells a share of stock when the option is exercised.
  - Expiration date: The last date on which an option holder has the right to exercise the option.
- **Put**: gives the owner the right, but not the obligation, to sell a fixed quantity of the underlying asset at the strike price at (or before) the expiration date.
- The price of an option or the option premium is paid by the buyer to the seller to acquire the right.

Terminology

- Exercising an option: when a holder of an option enforces the agreement and buys or sells a share of stock at the agreed-upon price.
- The option buyer (holder): holds the right to exercise the option and has a long position in the contract.
- The option seller (writer):
  - Sells (or writes) the option and has a short position in the contract.
  - Because the long side has the option to exercise, the short side has an obligation to fulfill the contract if it is exercised.

Terminology (con't)

- The intrinsic value of an option corresponds to the gain realised by the owner in the case of an immediate exercise.
  - At-the-money: describes an option whose exercise price is equal to the current stock price.
  - In-the-money: describes an option whose value, if immediately exercised, would be positive.
  - Out-of-the-money: describes an option whose value, if immediately exercised, would be negative.
  - Deep in-the-money (out-of-the-money): describes an option that is in-the-money (out-of-the-money) and for which the strike price and the stock price are very far apart.

Terminology (con't 2)

- **European option:** options that allow their holders to exercise the option only on the expiration date.
- **American option:** options that allow their holders to exercise the option on any date up to, and including, the expiration date.
  - Note: The names American and European have nothing to do with the location where the options are traded.
- **Asian option:** based on an average, e.g., the average monthly return of the CAC40 over the next five years.
- **Bermuda option:** can be exercised only on specific dates.

Rationale of option strategies

- Buyer (**long position**) of a **call** expects an increase in the price of the underlying.
- Seller (**short position**) of a **call** expects a decrease in the price of the underlying.
- Buyer (**long position**) of a **put** expects a decrease in the price of the underlying.
- Seller (**short position**) of a **put** expects an increase in the price of the underlying.

But first, let's fight a monster



The monster is called max

- In the next slides you will meet formulas like $\max(S_T - K;\ 0)$. That $\max$ is the whole monster.
- $\max$ of two numbers means: **look at both and keep the bigger one**. Nothing else.
  - $\max(3;\ 7) = 7$ and $\max(7;\ 3) = 7$: the order does not matter.
  - $\max(-5;\ 0) = 0$: when one of the two is zero, a negative number never wins.
- So $\max(\text{something};\ 0)$ is just a way of saying **"never let this go below zero"**.

Why an option needs it

- At expiration, the holder of the option **chooses** whether to use it. Nobody chooses to lose money, so the option is used only when it pays.
- Take a call with a strike price of $K = 20$, that is the right to buy the share for 20:
  - If the share is worth 25, you buy at 20 something worth 25: you gain 5.
  - If the share is worth 15, buying at 20 would be a bad deal, so you walk away and get 0.
- "Take the better of the two" is exactly what $\max$ writes down:

$$C_T = \max\left(S_T - K;\ 0\right)$$

The same thing in numbers

- Strike price $K = 20$. The share price at expiration is S_T:

Share price S_T	10	15	20	25	30
$S_T - K$	-10	-5	0	5	10
Call payoff $\max(S_T - K;\ 0)$	0	0	0	5	10
Put payoff $\max(K - S_T;\ 0)$	10	5	0	0	0

- The middle row can be negative, the payoff rows never are. The zero in the $\max$ does that work.

That is why the graph has an elbow

- The payoff diagrams you will see are just a picture of this choice:
  - On the flat part, $\max$ is picking the zero: the option is not worth using.
  - On the sloped part, $\max$ is picking $S_T - K$: the option is worth using, and every extra euro on the share is an extra euro of payoff.
- The two parts meet exactly at the strike price. That is the kink.

The max monster was kind of friendly after all



C. Option payoffs at expiration

- The value of a call at expiration is

$$C_T = \max\left(S_T - K;\ 0\right),$$

where S_T is the price of the underlying at expiration and K is the strike price of the option.

- Example for $K = 20$:

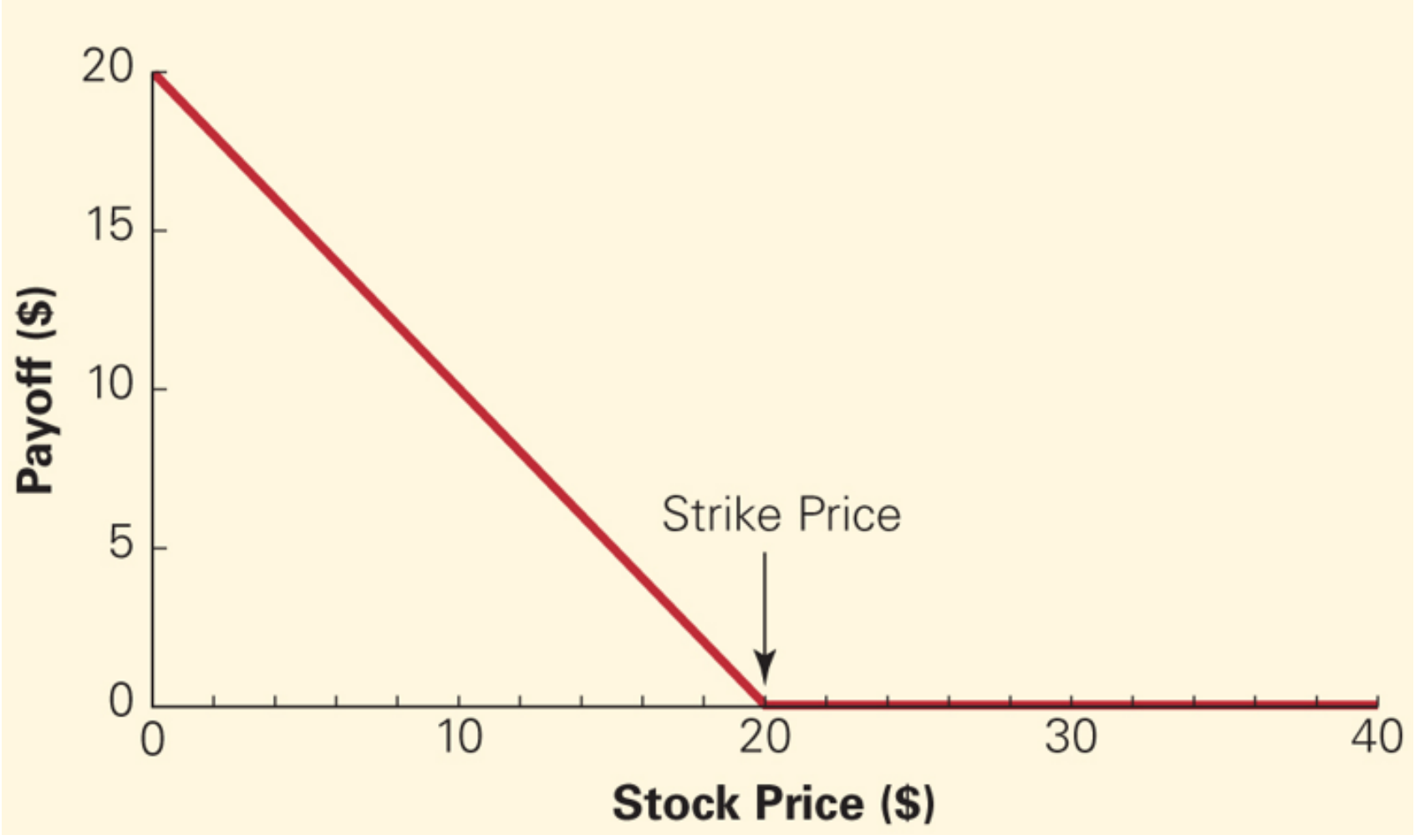


Option payoffs at expiration: put

- The value of a put at expiration is

$$P_T = \max\left(K - S_T;\ 0\right).$$

- Example for $K = 20$:



Option pricing model: call

- The most commonly used model to value European options is the Black-Scholes formula:

$$C_t = S_t N(d_1) - K e^{-r_f(T-t)} N(d_2)\,,$$

where

$$d_1 = \frac{ln\left(S_t/K\right) + \left(r_f + 0.5\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}},$$

$$d_2 = d_1 - \sigma\sqrt{T-t},$$

S_t is the current price of the underlying, $T-t$ is the time to expiration, K is the strike price, r_f is the risk-free rate, σ is the volatility, and $N(\cdot)$ is the cumulative distribution function of the standardized normal distribution.

- Remarks:
  - N(d_2) = "risk-neutral" probability that the call is "in the money" at expiration.
  - N(d_1) = "delta" of the option ($\Delta = tialC_t/tialS_t$).

Options: history

- Options have existed for a long time but they have gained in popularity since 1973. In that year:
  - the first regulated option market was introduced (the Chicago Board Options Exchange, or CBOE)
  - two academic articles pioneering option pricing were published.
- In Europe, the two main derivative exchanges are the NYSE Liffe and Eurex.
- A large number of options are not traded on exchanges but rather in over the counter markets.

Option quotes: example of CBOE

EBAY (EBAY INC) **33.75 −0.235**

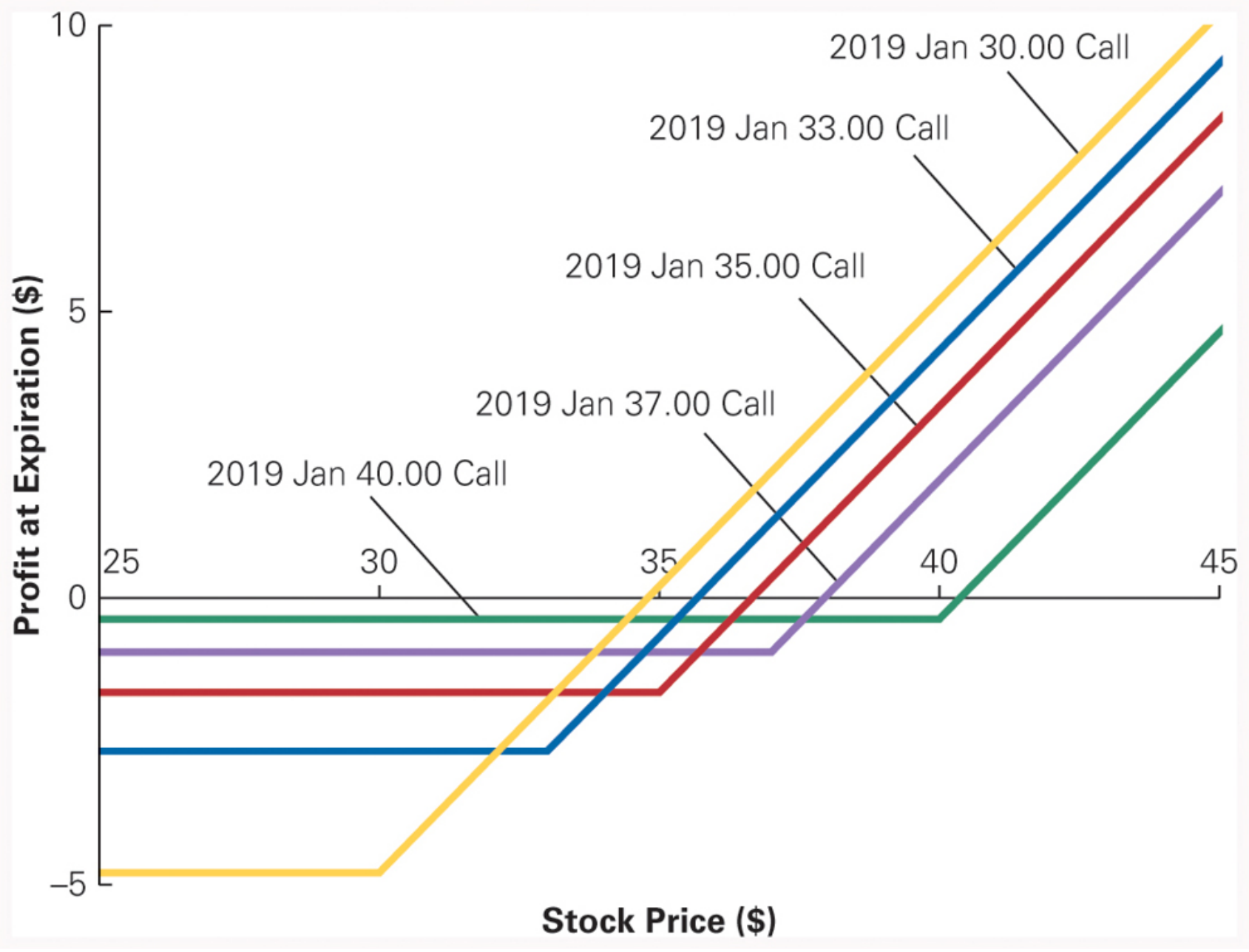
Sep 10 2018 @ 12:33 ET **Bid** 33.75 **Ask** 33.76 **Size** 18 x 14 **Vol** 2108011

Calls	Last Sale	Net	Bid	Ask	Vol	Open Int	Puts	Last Sale	Net	Bid	Ask	Vol	Open Int
2018 Sep 21 32.00 (EBAY1821I32)	1.86	–	1.86	1.96	–	534	2018 Sep 21 32.00 (EBAY1821U32)	0.09	−0.02	0.09	0.10	2	2122
2018 Sep 21 33.00 (EBAY1821I33)	1.04	–	1.03	1.08	15	827	2018 Sep 21 33.00 (EBAY1821U33)	0.25	−0.06	0.24	0.26	2	2927
2018 Sep 21 34.00 (EBAY1821I34)	0.45	−0.09	0.42	0.43	110	4959	2018 Sep 21 34.00 (EBAY1821U34)	0.62	0.03	0.63	0.65	35	5631
2018 Sep 21 35.00 (EBAY1821I35)	0.15	−0.06	0.14	0.15	258	7199	2018 Sep 21 35.00 (EBAY1821U35)	1.13	−0.22	1.34	1.39	1	3594
2018 Sep 21 36.00 (EBAY1821I36)	0.05	−0.03	0.05	0.06	268	13279	2018 Sep 21 36.00 (EBAY1821U36)	2.28	−0.07	2.15	2.30	2	1481
2019 Jan 18 30.00 (EBAY1918A30)	4.75	–	4.70	4.80	–	737	2019 Jan 18 30.00 (EBAY1918M30)	0.74	–	0.65	0.68	–	7912
2019 Jan 18 33.00 (EBAY1918A33)	2.70	−0.12	2.62	2.68	2	467	2019 Jan 18 33.00 (EBAY1918M33)	1.55	−0.05	1.55	1.58	129	7562
2019 Jan 18 35.00 (EBAY1918A35)	1.73	0.10	1.61	1.66	6	2079	2019 Jan 18 35.00 (EBAY1918M35)	2.47	–	2.51	2.57	–	15795
2019 Jan 18 37.00 (EBAY1918A37)	0.96	–	0.91	0.96	1	2324	2019 Jan 18 37.00 (EBAY1918M37)	3.80	0.65	3.85	3.90	79	17202
2019 Jan 18 40.00 (EBAY1918A40)	0.39	–	0.34	0.38	–	3455	2019 Jan 18 40.00 (EBAY1918M40)	5.60	–	6.30	6.40	–	6093

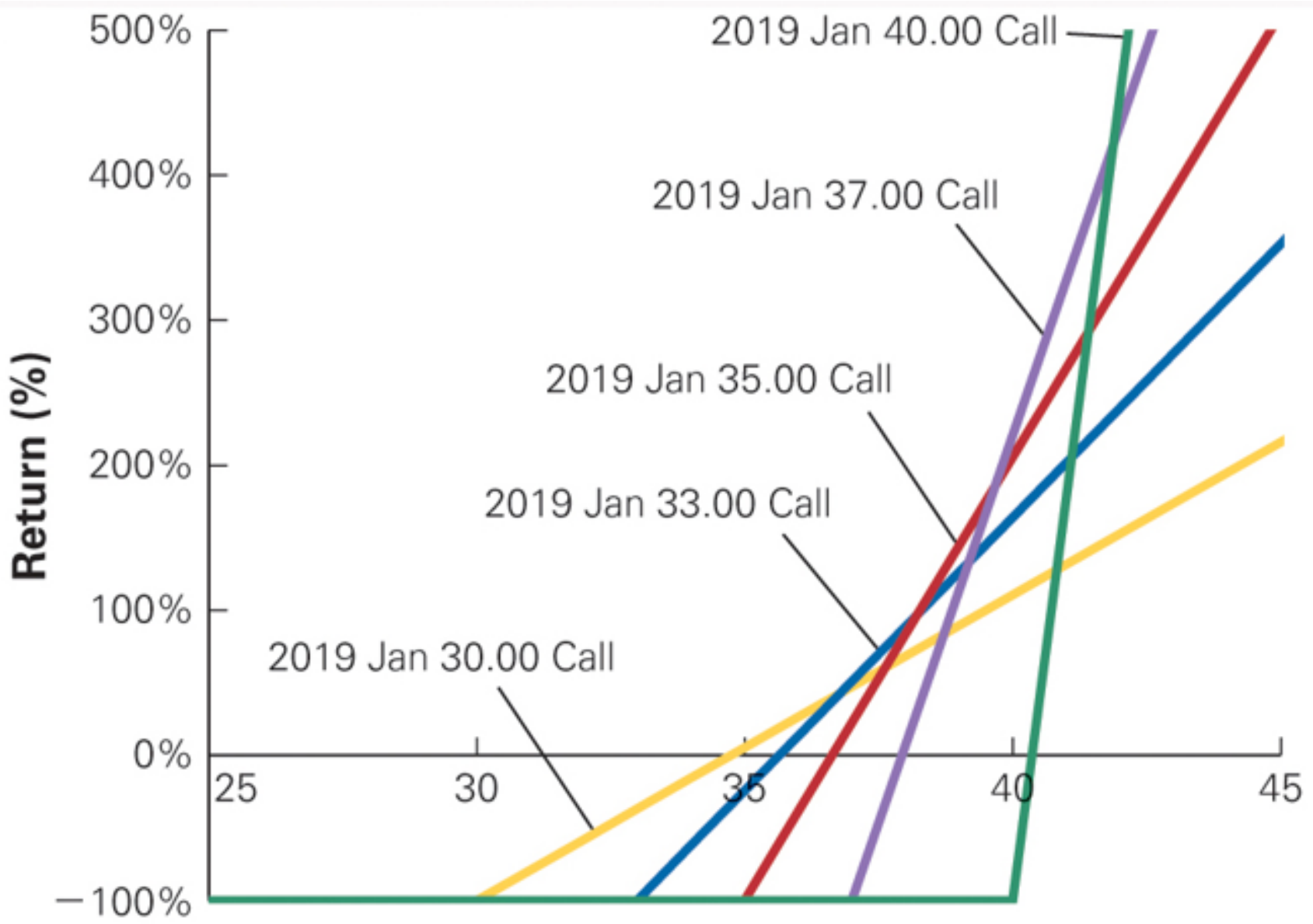
Suppose that you have decided to buy a call with a strike of 30 and expiration in January.

- Is this option “in-the-money” or “out-of-the-money”?
- From the table, compute the return (in %) obtained by an investor who chooses to buy the call today and hold it to expiry, assuming that the stock price reaches 40$ at the expiration of the call option.

Profit from holding a call option to expiration



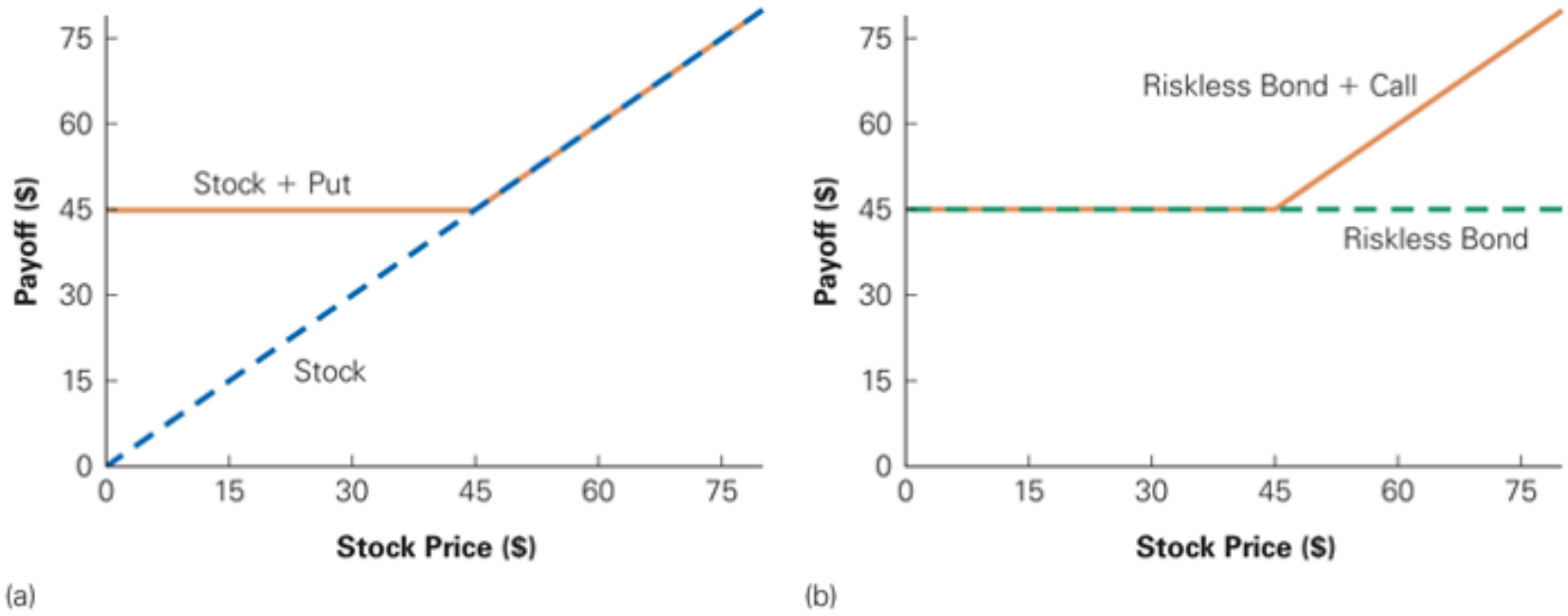
Returns from purchasing a call option and holding it to expiration



Option usage in practice: portfolio insurance

- Options can be used to protect a portfolio against a decline in its value.
- Consider a portfolio which only contains Amazon stocks. The investor is worried that the stock price may decline. At the same time, he is reluctant to sell his stocks because he also thinks there is a reasonable likelihood that the stock price will increase in the future.

Option usage in practice: portfolio insurance (con't)



The plots show two different ways to insure against the possibility of the price of Amazon stock falling below \$45:

(a) Orange line shows value on expiration date of a long position in Amazon stock and a European put option with a strike of \$45 (blue dashed line is the stock payoff).

(b) Orange line shows value on expiration date of a long position in a zero-coupon risk-free bond with a face value of \$45 and a

Quiz

- Does the holder of an option have to exercise it?
- Explain how you can use put options to create portfolio insurance. How can you create portfolio insurance using call options?